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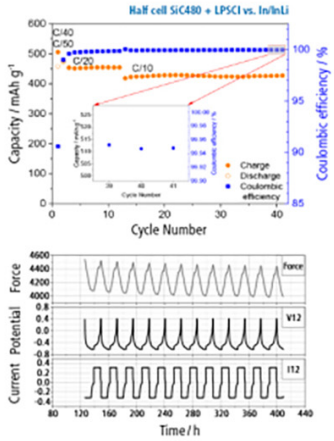

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JES Focus Issue on Mathematical Modeling of Electrochemical Systems at Multiple Scales in Honor of John Newman

Thermodynamic Model for Substitutional Materials: Application to Lithiated Graphite, Spinel Manganese Oxide, Iron Phosphate, and Layered Nickel-Manganese-Cobalt Oxide

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We derive and implement a method to describe the thermodynamics of electrode materials based on a substitutional lattice model. To assess the utility and generality of the method, we compare model results with experimental data for a variety of electrode materials: lithiated graphite and layered nickel-manganese-cobalt oxide (Chevrolet Bolt Electric Vehicle negative and positive electrode materials, respectively), manganese oxide (in the positive electrodes of the Gen 1 and Gen 2 Chevrolet Volt Extended Range Electric Vehicle and the positive electrode of many high-power-density batteries), and iron phosphate (Gen 1 Chevrolet Spark Electric Vehicle positive electrode material and of immediate interest for 12 and 48 V applications). An early version of the model has been applied to lithiated silicon (Li-Si). As was found in the Li-Si study, the model enables one to quantitatively represent experimental data from these different electrode materials with a small number of parameters, and, in this sense, the approach is both general and efficient. An open question is the utility of controlled-potential vs. controlled-current experiments for the elucidation of the system thermodynamics. We provide commentary on this question, and we highlight other open questions throughout this work.

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Central to the modeling of electrochemical systems, particularly batteries relying on solid-state diffusion within the active materials, is an accurate description of the system thermodynamics.[1–4] We describe and implement a thermodynamic model for substitutional electrode materials. At the heart of the model is a simple expression for the open-circuit potential U in terms of the fraction of filled sites x within a host material. To assess the value of the model, we apply it to four electrode materials of commercial interest today: lithiated graphite, spinel manganese oxide, iron phosphate, and nickel-manganese-cobalt oxide. The model is an expanded version of that described in Ref. 12. A slight alteration is needed to describe the nickel-manganese-cobalt oxide so that inaccessible lithium can be considered (giving rise to x_0 in Table I), and we also describe how reactions between sites within the electrode material can be addressed.

This document is organized as follows. First, we provide an overview of the materials and instrumentation for the experimental characterization of the electrode materials. An introduction of the thermodynamic model is then presented, followed by a detailed description of the model. A discussion of results is then presented, followed by a brief Appendix devoted to the convergence properties of the series summation employed in the $U(x)$ model.

Experimental

Both the graphitic carbon porous electrode and the $Ni_{0.6}Mn_{0.2}Co_{0.2}$ electrode are from cells used in the Chevrolet Bolt electric vehicle (cells supplied by LG Chem). LG Chem also provided the lithium iron phosphate electrode. Half cells were constructed with 1.59 cm diameter working-electrode (WE) disks and 2.70 cm diameter metallic lithium (Aldrich, 99.9%) counter-electrode (CE) disks. The cell was filled with electrolyte comprised of 1.0 M $LiPF_6$ in 50:50 by volume ethylene carbonate and diethyl carbonate (BASF) with < 20 ppm H_2O content. Separators were rinsed with HPLC-grade alcohol (90-5-5 vol.% ethanol, methanol, isopropanol, Alfa Aesar), then dried at 45°C under about 0.01 Torr vacuum. Other materials were dried at 100°C under vacuum of about 0.01 Torr. Cell assembly and cycling were performed under an inert argon atmosphere with O_2 and H_2O concentrations < 1 ppm. Cycling was performed with a Princeton Applied Research PMC 2000 potentiostat/galvanostat. For the voltammetry work on the $Ni_{0.6}Mn_{0.2}Co_{0.2}$ electrode, the experiments employed a VMP3 multichannel potentiostat (Biologic).

The lithiated spinel manganese oxide (LMO) electrodes were made using 88.5% LMO (Mitsui Mining), 3% PVDF binder (Kureha), and the remainder a mixture of conductive carbons KS6 (Timcal), Super P (Timcal), and vapor grown carbon fiber (Showa Denko). A 20 μm thick, 47–48% porous separator was employed (Celgard) between the lithium CE (Shanghai Oujin Co. Ltd.) and the WE. The electrolyte contained 1.0 M $LiPF_6$ (Do-Fluoride Chemicals Co. Ltd.), and a solvent phase of 15% propylene carbonate, 5% ethylene carbonate, 5% dimethyl carbonate, 5% diethyl carbonate, and 70% methyl butyrate (volume percents, Capchem). Electrochemical experiments employed a Solartron Analytical 1400.

For the reference electrode, a 50 μm diameter aluminum wire (Goodfellow, 99.99+%) was electrochemically alloyed in situ using Li from the CE to a target Li content of 25 atomic percent. The wire was located between two 25 μm thick separators with porosity of approximately 80% (Entek). Early history on the use of Li-Al reference electrodes can be found in Ref. 13. For Li contents between 20 and 30 atomic percent, a stable reference potential is obtained; hence, our target value of 25 atomic percent. The wire was alloyed at a current density of 50 $\mu A/cm^2$. The wire was then allowed to equilibrate for 48 h before use. In practice, potentials of the equilibrated Li-Al alloy wires varied between 0.375 and 0.385 V vs. Li/Li+. Because of side reactions during the lithiation, the composition cannot be expected to be exactly 25 atomic percent lithium, and this may account for the observed slight variations. Most important, the potential of

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Table I. Parameters for the thermodynamic model of the open-circuit potentials for the materials specified in the leftmost column. The Li-Si values are those reported in Ref. 12. For $Li_xNi_{0.6}Mn_{0.2}Co_{0.2}$, $x_0 \leq x \leq 1$, where $x_0 = 0.1325$, the fractional occupancy of Li at the maximum potential (vs. a Li reference) that yields stable cycling, 4.3 V.

Host	j	U_j^0	X_j	ω_j
Graphitic carbon $Li_x C_6$	1	0.08843	0.43336	0.08611
	2	0.12799	0.23963	0.08009
	3	0.14331	0.15018	0.72469
	4	0.16984	0.05462	2.53277
	5	0.21446	0.06744	0.09470
	6	0.36325	0.05476	5.97354
Silicon $Li_x Si_{4.4}$ Lithiation (reduction)	1	0.08216	0.21834	0.70934
	2	0.23290	0.23264	0.95970
	3	0.12606	0.42492	3.02803
	4	0.42638	0.12410	4.68406
Silicon $Li_x Si_{4.4}$ Delithiation (oxidation)	1	0.28070	0.24314	1.11628
	2	0.47931	0.05008	0.88258
	3	0.45049	0.38526	3.07196
	4	0.71796	0.17304	4.27128
Iron phosphate $Li_x FePO_4$	1	3.42977	0.82584	0.06098
	2	3.42977	0.17416	0.35868
Manganese oxide (spinel) $Li_x Mn_2O_4$	1	4.01326	0.55286	1.46502
	2	4.14834	0.39411	1.06156
	3	4.14450	0.05303	0.39293
	1	4.01173	0.55070	1.52000
	2	4.14902	0.44930	0.93000
$Li_x Ni_{0.6}Mn_{0.2}Co_{0.2}$ $x_0 = 0.1325$	1	3.62274	0.13442	0.96710
	2	3.72645	0.32460	1.39712
	3	3.90575	0.21118	3.50500
	4	4.22955	0.32980	5.52757

the wire reference remained fixed within 1 to 2 millivolts over a week, yielding more than sufficient stability for the purposes of this work.

Theory for the Equilibrium Thermodynamics

How do we identify an electrochemical reaction?—Current peaks in voltammograms are one way electrochemical reactions can be identified.[5] However, for commonly used equipment to characterize batteries, such as that used here and referenced in the Experimental section of this work, higher signal-to-noise is obtained at low rates using constant-current operation. Perhaps for this reason, constant-current experiments are more commonly seen in the battery literature, despite the fact that evidence for reactions in the measured potential trace is obscured relative to current peaks in voltammograms. The similarities and differences associated with constant-current operation and linear-sweep voltammetry will be highlighted in this introductory section. We restrict our attention to low rates of operation and "dynamic equilibrium" (i.e., no measureable irreversible behavior).

Consider the elementary reaction

$$O + e^- \rightleftarrows R \qquad [1]$$

where O is the oxidized form and R is the reduced form of the reactant. The Nernst equation represents the ideal open-circuit potential (ocp or U) of the working electrode relative to a reference electrode:[2,5]

$$U = U^0 + \frac{RT}{F} \ln \frac{c_O}{c_R} \qquad [2]$$

Faraday's constant, the universal gas constant, and absolute temperature, are given by F, R, and T, respectively, and c denotes concentration. One implication of the Nernst equation is that the a Boltzmann distribution describes the concentration ratio, $\frac{c_O}{c_R} = e^{\frac{F}{RT}(U-U^0)}$. For real systems of practical interest, it is usually necessary to use activity coefficients[2,5] instead of concentrations. For the purposes of this work, we employ a close variant of Eq. 2:

$$U = U^0 + \omega \frac{RT}{F} \ln \frac{x_O}{x_R}$$

$$U(x) = U^0 + \omega \frac{RT}{F} \ln \frac{1-x}{x} \text{ or } x(U) = \frac{1}{1 + e^{f(U-U^0)/\omega}} \text{ and}$$

$$\frac{dx}{dU} = -\frac{1}{\omega} \frac{f e^{f(U-U^0)/\omega}}{\left(1 + e^{f(U-U^0)/\omega}\right)^2} \qquad [3]$$

where $f = F/(RT)$ and ω is introduced as an adjustable parameter; we will show in the next section that ω can be related directly to the appropriate activity coefficients. This approach yields a modified Boltzmann distribution, $\frac{c_O}{c_R} = e^{\frac{F}{\omega RT}(U-U^0)}$. The success (or failure) of this model may be assessed by its utility (or lack thereof) in representing different electrode materials of interest, and that is a primary aspect of this work. In writing Eq. 3, we have assumed a constant total concentration $c_T = c_O + c_R$, allowing for the use of mole fractions, and, to streamline exposition, we use the notation $x = x_R = 1 - x_O$. (We employ mole fractions as we are interested in insertion electrodes in this work, and the mole fraction is equal to the fractional occupancy of available sites in the host material.)

The current density i corresponds to the flow of charge Q per unit cross-sectional area with time t and $x = Q/Q_{\max}$; hence,

$$i = \frac{dQ}{dt} = Q_{\max} \frac{dx}{dt} \qquad [4]$$

In the dynamic equilibrium condition, the charge Q is a function of the potential U, as is x, and vice versa. Thus, by use of the chain-rule,

$$i = Q_{\max} \frac{dU}{dt} \frac{dx}{dU} = Q_{\max} \nu \frac{dx}{dU} \text{ or } \frac{dx}{dU} = \frac{i}{\nu Q_{\max}} \qquad [5]$$

where ν is the potential sweep rate. From this equation, we see that the current i is a function of U, as dx/dU can be calculated by means of Eq. 3. The impact of ω on $U(x)$ and dx/dU is displayed in Figure 1 for the elementary reaction of Eq. 1. Peaks, corresponding to maxima in $(dx/dU)/\min(dx/dU)$, are centered about the standard potential $U^0 = 0.4$ V, and the peaks in the dx/dU curves can be immediately associated with a reaction—hence, in answer to the question posed at the beginning of this section, peaks in a plot of dx/dU vs. U can be used to identify an electrochemical reaction. The mathematical equivalence of dx/dU vs. U and linear-sweep voltammetry analysis is clarified by the plots shown in Figure 2, where results for two reactions are depicted, and dx/dU (left plot) and current peaks (right plot) are centered around standard electrode potentials of the two reactions (0.4 and 0.6 V) for constant-current operation and linear-sweep voltammetry.

The importance of Eq. 5 is that dx/dU can be obtained by linear-sweep voltammetry without the need for numerical differentiation of data, as is needed to deduce dx/dU from constant-current results. However, as mentioned briefly above, modern-day experimental equipment used to characterize battery electrodes has difficulty in providing high-quality current-potential data for very slow potential sweep rates (e.g., below 0.0025 mV/s in this work), and this limitation can favor the use of constant-current operation to identify reactions by means of dx/dU vs. U plots, as is done in much of this work, instead of using linear-sweep voltammetry. In addition, for systems with near two-phase behavior (e.g., lithiated graphite and iron phosphate), low potential

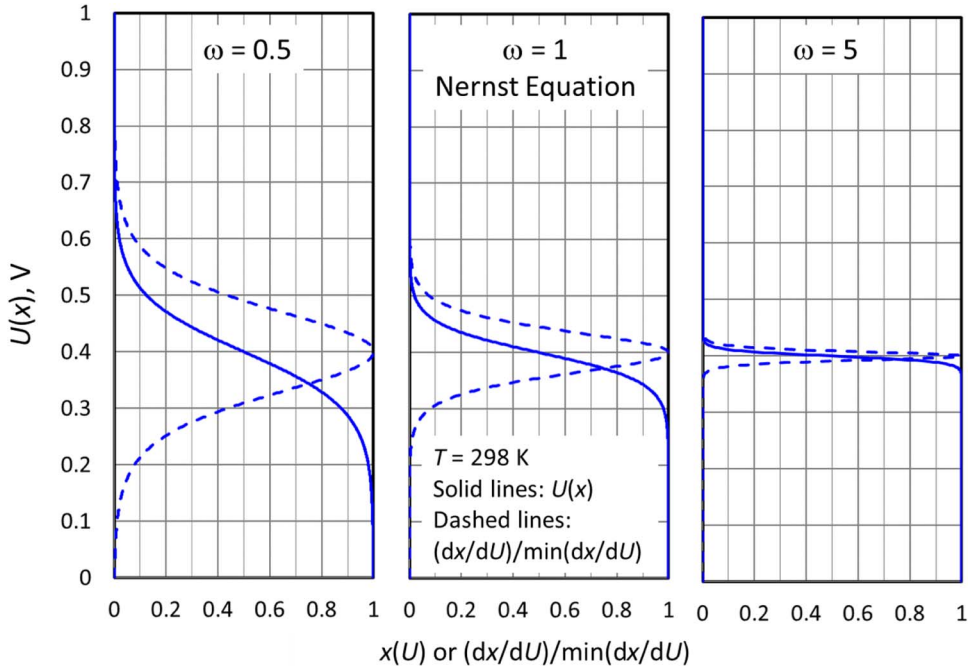

Figure 1. Impact of the parameter ω on the potential profile $U(x)$ for a single-reaction system (cf. Eq. 3, $U^0 = 0.4$ V).

sweep rates through potential plateaus in the U-x plane can yield unwanted high rates of reaction and significant irreversible losses; in such cases, dynamic equilibrium no longer prevails.

Thermodynamics for multi-reaction, multi-species systems.— While the treatment provided will be general for substitutional materials, we couch the exposition in terms of lithium insertion electrodes. Elementary electrochemical reactions at the particle surface are represented by the following reaction

$$\mathrm{Li}^+_{liquid} + e^-_{solid} + H_{j,solid} \underset{k_{a,j}}{\overset{k_{c,j}}{\rightleftarrows}} (\mathrm{Li}{-}H_j)_{solid} \qquad [6]$$

This equation states that for each reaction j, a host site H_j can accommodate one Li, leading to the filled site $\mathrm{Li}{-}H_j$. The anodic and cathodic rate constants for reaction j are given by $k_{a,j}$ and $k_{c,j}$, respectively. At the Li reference electrode,

$$\mathrm{Li}^+_{liquid} + e^-_{solid} \rightleftarrows \mathrm{Li}_{solid} \qquad [7]$$

For now, we assume the electrolyte solution affords no resistance, and the concentration of Li^+_{liquid} is uniform. Equilibrium is characterized by

$$\begin{aligned} -FU &= \Delta G_j = \mu_j - \mu_{H,j} - \mu_{\mathrm{Li}} \\ &= \mu^0_j - \mu^0_{H,j} - \mu^0_{\mathrm{Li}} + RT \ln \frac{a_j}{a_{H,j}} \\ &= \Delta G^0_j + RT \ln \frac{a_j}{a_{H,j}} \\ FU &= FU^0_j + RT \ln \frac{a_{H,j}}{a_j} \end{aligned} \qquad [8]$$

where ΔG, U, μ, and a represent the Gibbs free energy of reaction, electrode potential, chemical potential ($\mu_j$being the chemical

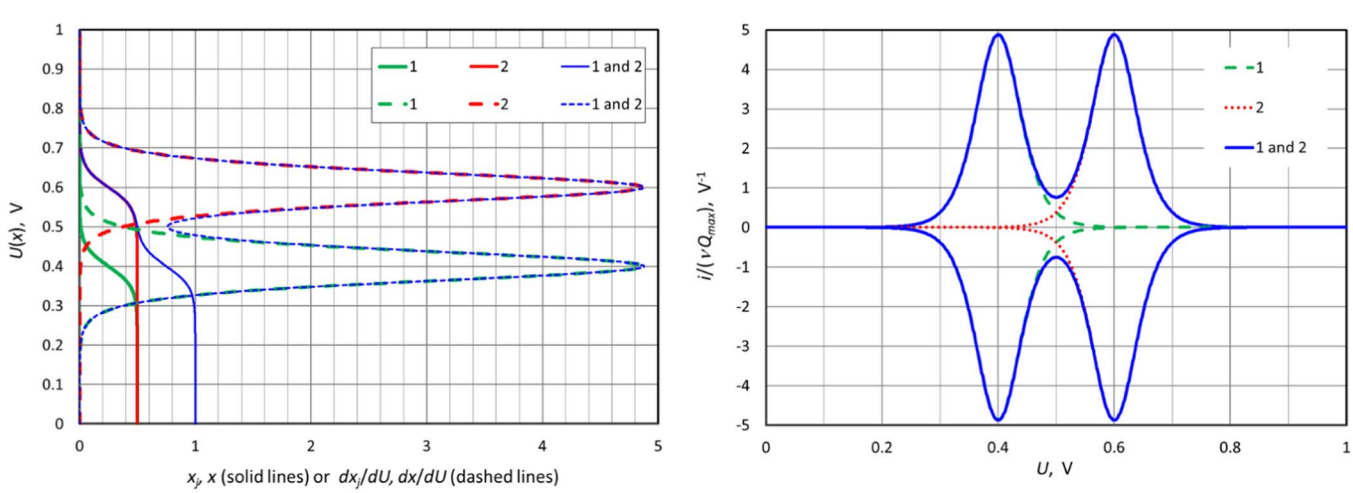

Figure 2. Potentials vs. mole fractions and derivatives of the same (left plot) and voltammogram (right plot). (See Eqs. 3 and 5.) For these plots, $\omega_1 = \omega_2 = 1$. Peaks in both plots are seen about $U^0_1 = 0.4$ V and $U^0_2 = 0.6$ V.

potential of a filled site in gallery j, $\mu_j = \mu_j^0 + RT \ln a_j$, and $\mu_{H,j}$ being the chemical potential for a vacant site in gallery j, $\mu_{H,j} = \mu_{H,j}^0 + RT \ln a_{H,j}$), and activity (same subscript convention). The superscript 0 refers to standard conditions.

We shall expand on the approach first described in Ref. 6 for graphite, and later used in Ref. 7 (again, Li-graphite) and 8 for Li-Ga. Birkl et al.[9] employed a similar approach (Eq. 9 below) to obtain parameter-fit expressions for full-cell open-circuit potentials of commercial lithiated graphite/metal oxide batteries. Researchers at Argonne National Laboratory have also employed Eq. 9 below for partially graphitic[10] and metal oxide[11] electrodes. Further discussion of the motivation and utility of this approach can be found in Ref. 12 for the Li-Si system. Similar to Eq. 3, the open circuit potential is taken to be:

$$U_j(x_j) = U_j^0 + \frac{\omega_j}{f} \ln\left(\frac{x_{H,j}}{x_j}\right) = U_j^0 + \frac{\omega_j}{f} \ln\left(\frac{X_j - x_j}{x_j}\right) \quad [9]$$

where $f = F/(RT)$. Under equilibrium conditions, $U = U_1 = U_2 = \ldots = U_j$, where U is the electrode potential relative to a Li reference located in solution (cf. Eq. 7), and x_j and $x_{H,j} = (X_j - x_j)$ represent the fraction of sites j filled with Li, $(\mathrm{Li} - H_j)_{solid}$, and the fraction of vacant host sites j, $H_{j,solid}$, respectively. The fraction of sites available for Li that are associated with a species and a reaction j is given by X_j. The electrode potential uniquely determines the composition (site occupancy) of each collection of sites j, which we shall refer to as galleries, and species x_j, as can be seen by inverting Eq. 9:

$$x_j(U) = \frac{X_j}{1 + \exp\left[f(U - U_j^0)/\omega_j\right]} \text{ and } 0 < x_j < X_j \quad [10]$$

Site conservation leads to an expression for the total fraction (fractional occupancy) of Li-filled sites x:

$$x(U) = \sum_j x_j(U) = \sum_j \frac{X_j}{1 + \exp\left[f(U - U_j^0)/\omega_j\right]} \text{ and}$$
$$0 < x(U) < \sum_j X_j \quad [11]$$

For the four electrode materials investigated in this work, $\sum_j X_j = 1$. From Eqs. 10 and 11,

$$\frac{dx_j}{dU} = -\frac{X_j}{\omega_j} \frac{f \exp\left[f(U - U_j^0)/\omega_j\right]}{\left\{1 + \exp\left[f(U - U_j^0)/\omega_j\right]\right\}^2} \text{ and } \frac{dx}{dU} = \sum_j \frac{dx_j}{dU} \quad [12]$$

Equation 8 and 9 can be used to identify activity coefficients γ:[12]

$$FU_j = FU_j^0 + \omega_j RT \ln\left(\frac{x_{H,j}}{x_j}\right) = FU_j^0 + RT \ln\left(\frac{\gamma_{H,j} x_{H,j}}{\gamma_j x_j}\right)$$
$$= FU_j^0 + RT \ln\left(\frac{a_{H,j}}{a_j}\right) \quad [13]$$

where

$$\gamma_j = (x_j)^{\omega_j - 1} \qquad a_j = (x_j)^{\omega_j}$$
$$\gamma_{H,j} = (x_{H,j})^{\omega_j - 1} = (X_j - x_j)^{\omega_j - 1} \qquad a_{H,j} = (x_{H,j})^{\omega_j} = (X_j - x_j)^{\omega_j} \quad [14]$$

and

$$U_j^0 = \frac{1}{f} \ln \frac{k_{c,j}}{k_{a,j}} \quad [15]$$

From this formulation, it is clear that the influence of the activity coefficients vanishes when $\omega_j = 1$, resulting in unit activity coefficients, and the Nernst equation is recovered, corresponding to an ideal, single phase system. In contrast, as ω_j approaches zero, two-phase (constant potential) behavior is obtained.

We shall now allow exchange of Li between different galleries. The approach complements that employed in Ref. 11. Specifically, we allow Li to be exchanged in accordance with the reactions

$$\left.(\mathrm{Li}{-}H_k)_{solid} + H_{j,solid} \underset{k_{b,j-k}}{\overset{k_{f,j-k}}{\rightleftarrows}} H_{k,solid} + (\mathrm{Li}{-}H_j)_{solid}\right\}$$
$$\frac{1}{2} J(J-1) \text{ equations} \quad [16]$$

For each $j{-}k$ Li-exchange reaction, we can express the Gibbs free energy ΔG_{j-k} as

$$\Delta G_{j-k} = \mu_j + \mu_{H,k} - \mu_k - \mu_{H,j}$$
$$= \mu_j^0 + \mu_{H,k}^0 - \mu_k^0 - \mu_{H,j}^0 + RT \ln \frac{a_j a_{H,k}}{a_k a_{H,j}} \quad [17]$$
$$= \Delta G_{j-k}^0 + RT \ln \frac{a_j a_{H,k}}{a_k a_{H,j}}$$

and

$$\Delta G_{j-k}^0 = RT \ln \frac{k_{f,j-k}}{k_{b,j-k}} \quad [18]$$

We can relate the free energies ΔG_{j-k} and ΔG_{j-k}^0 to potentials U_j and U_j^0 by substituting Eq. 13 into Eq. 17:

$$\Delta G_{j-k} = -F(U_j - U_k) \text{ and } \Delta G_{j-k}^0 = -F(U_j^0 - U_k^0) \quad [19]$$

At equilibrium, all U_j are equal to U, and, hence, all $\Delta G_{j\text{-}k}$ vanish. Thus, the addition of site-exchange reactions (Eq. 16) does not alter the equilibrium concentrations.

For diffusion phenomena, we will need (i) to calculate the Darken factor, $1 + \frac{d \ln \gamma_j}{d \ln x_j}$, and (ii) an understanding of the Gibbs-Duhem equation.[7,12] The Gibbs-Duhem equation (at uniform and constant temperature, pressure, stress, and other intensive variables) can be written as

$$\sum_{j=1}^{J} \left[x_j d\mu_j + x_{H,j} d\mu_{H,j}\right] = \sum_{j=1}^{J} \left[x_j d\mu_j + (X_j - x_j) d\mu_{H,j}\right] = 0 \quad [20]$$

As mentioned in the context of Eq. 8, $\mu_j = \mu_j^0 + RT \ln a_j$ and $\mu_{H,j} = \mu_{H,j}^0 + RT \ln a_{H,j}$, which, after eliminating the activities by means of Eq. 14, yields $d\mu_j = \omega_j RT/x_j$ and $d\mu_{H,j} = -\omega_j RT/(X_j - x_j)$. These relations can be used to show that

$$x_j d\mu_j + (X_j - x_j) d\mu_{H,j} = 0 \quad [21]$$

which demonstrates that the Gibbs-Duhem equation can be applied to each gallery (i.e., the galleries are de-coupled) without further approximation. Furthermore, Eqs. 8, 9, and 21 can be combined to show that

$$\omega_j = 1 + \frac{d \ln \gamma_j}{d \ln x_j} = -\frac{x_j(X_j - x_j)}{X_j} \frac{F}{RT} \frac{dU}{dx_j} \quad [22]$$

which is a constant that must be greater the zero, as $\frac{dU}{dx_j}$ is monotonic and negative for energy-storage materials.

Additional insight into the thermodynamic model can be deduced by examining the free energy of a gallery. We can express the free energy $G_{j-H,j}$ for each gallery containing filled and unfilled sites of composition x_j and $x_{H,j} = X_j - x_j$, respectively, as[14,15]

$$G_{j-H,j} = x_j \mu_j + x_{H,j} \mu_{H,j}$$
$$= x_j(\mu_j{}^0 + RT \ln a_j) + x_{H,j}(\mu_{H,j}^0 + RT \ln a_{H,j}) \quad [23]$$
$$= x_j(\mu_j{}^0 + \omega_j RT \ln x_j) + x_{H,j}(\mu_{H,j}^0 + \omega_j RT \ln x_{H,j})$$

where Eq. 14 has been employed for the activity of each species. Note that the free energy of a gallery $G_{j-H,j}$ is not to be confused the free energy of a reaction (ΔG_j in Eq. 8 or ΔG_{j-k} in Eq. 19). The

corresponding ideal free energy of a gallery is given by[14,15,d]

$$G^{Ideal}_{j-H,j} = x_j(\mu_j{}^0 + RT\ln x_j) + x_{H,j}(\mu^0_{H,j} + RT\ln x_{H,j}) \quad [24]$$

Combining Eqs. 23 and 24, we can express the free energy of each gallery as

$$G_{j-H,j} = \omega_j G^{Ideal}_{j-H,j} + (1-\omega_j)\left(x_j\mu_j{}^0 + x_{H,j}\mu^0_{H,j}\right)$$
$$\begin{cases}\omega_j = 1 \; G_{j-H,j} = G^{Ideal}_{j-H,j} & \text{Ideal, single phase}\\ \omega_j = 0 \; G_{j-H,j} = x_j\mu_j{}^0 + x_{H,j}\mu^0_{H,j} & \text{Two phase}\end{cases} \quad [25]$$

which underscores that the variation of ω_j can be used to capture behavior characteristic of single-phase and two-phase systems, consistent with the discussion immediately below Eq. 15. When $\omega_j = 1$, the gallery is an ideal, single-phase, binary system. When $\omega_j = 0$, the gallery is a two-phase system, with the free energy of the gallery dependent only on the chemical potentials of the two pure species, $\mu_j{}^0$ and $\mu^0_{H,j}$, and no compositional dependence is seen.

In the Appendix, we address convergence of the series if Eq. 11 in terms of representing $U(x)$ curves.

Some comments on the notions of multiple species and multiple reactions.—It has already been noted that current peaks in voltammograms are one way in which multiple electrochemical reactions can be identified. It is further noted in the Discussion section below that the multiple species that arise in the above formalism are generally consistent with results from X-ray diffraction. These observations supply motivation for the approach taken in this work. Although we do not pursue these details here, multiple reactions require multiple exchange current densities to characterize them, as done in Ref. 12, and multi-species transport entails multiple diffusion coefficients such as, for example, the binary diffusion coefficients appearing in the Stefan-Maxwell equations.[2,16] The precise nature of lithium transport in host materials is still the subject of ongoing research, but in Ref. 7 it was observed that diffusion in graphite associated with PITT measurements in Ref. 17 could be well described using a Darken factor with only one fit diffusion constant; that is, good agreement with experimental data was obtained with a single lithium species in the graphite host, along with its associated vacancies. We are thus led to ask if the thermodynamic formalism presented above can be used to describe the kinetics of multiple reactions while at the same time describing transport associated with only one lithium species. In this section, we describe how this can be done.

The key additional assumption that must be made is that the reactions 16 between galleries are always in equilibrium. From Eq. 19, this implies that $U_1 = U_2 = ... = U_J = U(x)$ for all J species, where we use the function $U(x)$ to denote this common potential, which depends only on the total mole fraction of lithium. From Eq. 10, it now follows that each species $x_j = x_j(U)$, so that the individual mole fractions can now be determined from knowledge of the total mole fraction and the function $U(x)$.

We now define total chemical potentials $\mu(x)$ and $\mu_H(x)$ using the equations

$$\mu dx = \sum_j \mu_j dx_j, \quad \mu = \sum_j \mu_j \frac{dx_j}{dx}, \quad \mu_H = \sum_j \mu_{H,j}\frac{dx_{H,j}}{dx_H}$$
$$x_j + x_{H,j} = X_j, \quad x + x_H = 1, \quad \frac{dx_j}{dx} = \frac{\frac{dx_j}{dU}}{\frac{dx}{dU}} \quad [26]$$
$$dx_j = -dx_{H,j}, \quad dx = -dx_H, \quad \mu_H = \sum_j \mu_{H,j}\frac{dx_j}{dx}$$

[d]The excess free energy of a gallery is given by $G^{Excess}_{j-H,j} = G_{j-H,j} - G^{Ideal}_{j-H,j}$. One can use the procedures outlined in Refs. 14 and 15 to determine activity coefficients from the excess free energy, with the result corresponding to the expressions in Eq. 14.

It then follows that

$$\mu - \mu_H = \sum_j \left(\mu_j - \mu_{H,j}\right)\frac{dx_j}{dx}$$
$$-FU = \mu_j - \mu_{H,j} - \mu_{Li}$$
$$\mu - \mu_H = \sum_j (\mu_{Li} - FU)\frac{dx_j}{dx}$$
$$= (\mu_{Li} - FU)\sum_j \frac{dx_j}{dx} = (\mu_{Li} - FU)$$
$$-FU = \mu - \mu_H - \mu_{Li} \quad [27]$$

The last of the above equations is needed to relate the potential $U(x)$ to the total chemical potentials for occupied and vacant sites $\mu\ (x)$ and $\mu_H(x)$. At constant temperature, pressure, and zero stress (i.e., no variations in any relevant intensive variables other than concentration), the differential of the Gibbs free energy and the Gibbs-Duhem equation take the form

$$\sum_j \mu_j dx_j + \mu_{H,j}dx_{H,j} = dG = \mu dx + \mu_H dx_H$$
$$x d\mu + x_H d\mu_H = 0 = x d\mu + (1-x)\,d\mu_H \quad [28]$$
$$d\mu_H = -\frac{x}{(1-x)}d\mu$$

It then follows that[7,16]

$$-FdU = d\mu - d\mu_H = \frac{1}{(1-x)}d\mu$$
$$\nabla\mu = -F(1-x)\frac{dU}{dx}\nabla x \quad [29]$$

This last equation is what is needed to write a single diffusion equation based on chemical potential or the open-circuit voltage $U(x)$.

The virtue of the above formalism is that one has the choice of using the individual species x_j when formulating kinetics equations, but one can also use the total mole fraction x when formulating a transport equation. It should be noted that, even if the inter-species reactions of Eq. 16 are equilibrated under low signal excitation, such as in EIS and PITT experiments, the equilibrium conditions could cease to hold under high current conditions. In this case, a more general form of transport might be needed, involving multiple diffusion coefficients and species, and kinetic resistance may need to be considered for the reactions represented by Eq. 16.

Discussion of Results

For all of the electrode materials investigated, Eq. 10 was used to calculate $x_j(U)$, Eq. 11 for $x(U)$, and Eq. 12 for dx_j/dU and dx/dU. The parameters U^0_j, X_j, and ω_j were fit by minimizing the least-squares difference between the experimental data and the model calculations as described in Ref. 12. Table I provides a listing of the fit parameters for the electrode materials analyzed in this work as well as those of Li-Si from Reference 12. While a similar model was employed to describe Li-Si, Li-Si exhibits a large hysteresis, which necessitates treating the oxidation (delithiation) and reduction (lithiation) curves separately. In Ref. 18, a brief review of hysteresis phenomena is provided, and an experimental procedure and a theoretical treatment are provided for describing Li-Si hysteresis. For the four materials investigated in this work, the observed difference between the oxidation and reduction curves measured at low rates is deemed to be small enough to ignore hysteresis. We now describe results for the materials characterized in this work.

Graphitic carbon.—Much has been written about lithiated graphite.[6,7,19–31] The material is central to the development and commercialization of lithium ion batteries used today, and the staging behavior of Li_xC_6 ($0 \le x \le 1$) is demonstrative of the essential features of intercalation materials. Shown in Figure 3 are the experimental

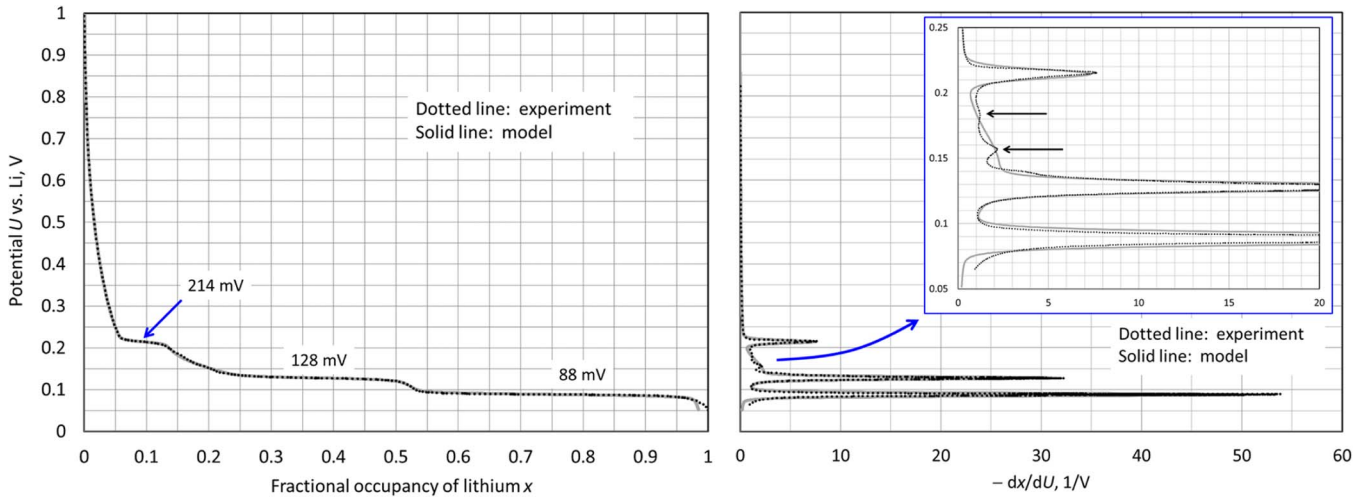

Figure 3. Model-experiment comparison for the open-circuit potential (left) and differential voltage spectroscopy (right) for lithiated graphite. The data were taken at approximately a C/100 rate.

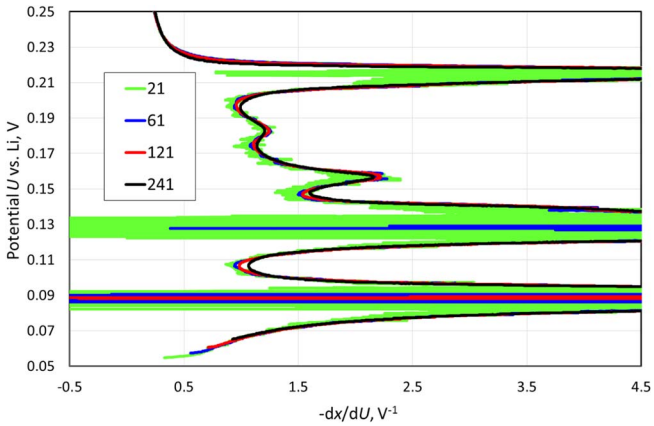

Figure 4. Expanded view of the experimental data for differential voltage spectroscopy plot of Figure 1. A least-squares fit of the slope in the $U(x)$ curve, using the indicated number of data points shown in the legend was employed to obtain the dx/dU curves. The 241-data-point curve is the same as that shown in Figure 3.

data for $U(x)$, or, equivalently, $x(U)$, and dx/dU. The model-experiment comparisons are quantitative insofar as the difference between the two are less than those associated with experiment-to-experiment variation and are difficult to discern in the $U(x)$ plot that spans the minimum and maximum practical values of U and x. (We shall employ this definition of quantitative in model-experiment comparisons to follow.)

Because we examine constant-current experiments, as previously mentioned, we must differentiate the $U(x)$ curves to obtain the dx/dU curves. For the left plot of Figure 3, all 14,333 data points are shown. To get the right plot that employs dx/dU, a least-squares fit of the slope to the $U(x)$ curve was obtained,[e] centered about the plotted value of U, using the indicated number of data points shown in the legend of Figure 4. Too few data points in the estimation of dx/dU yields too much noise to be useful, while the use of too many data points leads to excessive data loss. The effects of data loss can be seen by comparing, for example, the 241 and 121 data-point curves in Figure 4 near the peaks and valleys. When more points are employed, the sharpness of the peaks and valleys are reduced due to data averaging. Of the curves shown in Figure 4, only the curve corresponding to 241 data points did not yield negative abscissa values near 90 mV; negative values would indicate that U increases as x increases, which is not reasonable. The 241-data-point curve is the same as that shown in right plot of Figure 3.

Highlighted with arrows within the inset plot of Figure 3 are two small peaks (reactions) within the dx_j/dU vs. U plot (at about 0.158 and 0.183 V vs. Li). They are consistently obtained for the graphitic material of this study, and we have not seen them reported previously. Because the capacity associated with each peak is quite small relative to the electrode's total capacity, we shall ignore them in the modeling of the system, and we shall not address them further.

To clarify salient aspects of the model, calculations for $x_j(U)$ and dx_j/dU are provided in Figure 5, and activities for the various Li species j of lithiated graphite are plotted in Figure 6. For the three near-two-phase regions (Table I, $\omega_j < 0.1$, corresponding galleries $j = 1$, 2, and 5), the $x_j(U)$ profiles are nearly constant, consistent with Gibbs' phase rule. The other species profiles and reactions show more variation in $x_j(U)$, consistent with greater single-phase behavior. The activity coefficient plots in Figure 6 demonstrate the symmetry associated with the power-law expressions for the activity coefficients (Eq. 14), yielding symmetry for $a_j - x_j$ and $a_{H,j} - x_j$ curves with respect to one another. The fit values for the standard electrode potential U_j^0 and the fraction of available sites X_j associated with each

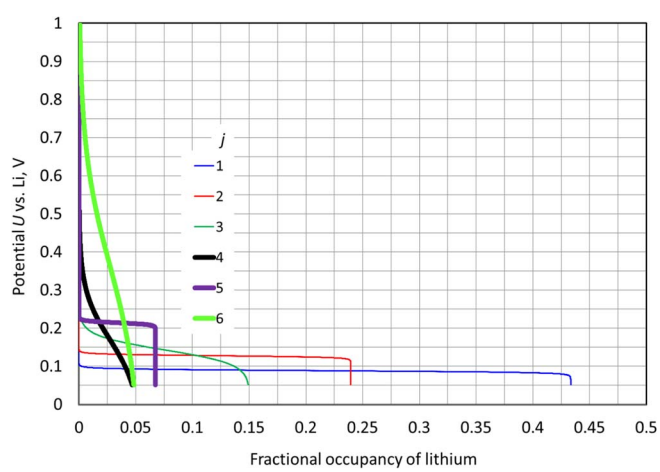

Figure 5. Open-circuit potential for the various reactions j of lithiated graphite. Reactions 1, 2, and 5 have two-phase character and substantially flat ocp's.

[e]The least-squares fit to determine dx/dU was obtained in this work by using the Slope function in Excel, as described at http://www.excelfunctions.net/Excel-Slope-Function.html.

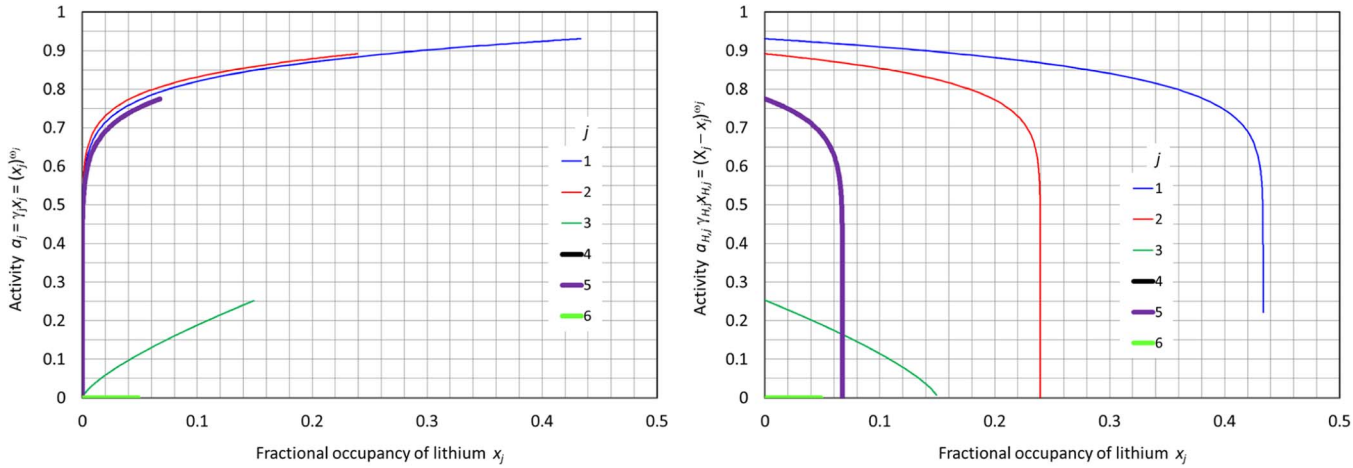

Figure 6. Activities for the various Li species j of lithiated graphite. The curves for species 4 lie below those of species 6 as plotted.

reaction of potential U_j^0 are in good agreement with X-ray analysis (e.g., see Ref. 19) for $j = 1$ to 4. Less is known about the higher stages ($j = 5$ and 6), comprising about 10% of the Li sites (i.e., as can be seen from Table I, $X_5 + X_6 = 0.12$).

Spinel manganese oxide.—One of the more well-studied and useful positive-electrode materials is spinel manganese oxide,[32–40] $Li_xMn_2O_4$ (or LMO), with $0 \leq x \leq 1$ for the higher voltage plateaus that provide stable cycling. Helpful reviews of various positive electrodes for lithium-based batteries can be found in Refs. 41–44. A recent review of LMO is provided in Ref. 46. Prior to the development of lithium ion batteries employing graphitic negative electrodes, companies worked to commercialize cells with lithium metal negatives and spinel manganese oxide positives. Relative to other positive-electrode materials, spinel manganese oxide admits high rates of lithium diffusion, consistent with the needs of high-power batteries, which, in addition to its low cost, is why it is still used in volume today, despite its lower Coulombic capacity.

Unlike the graphitic carbon analysis, we employed linear-sweep voltammetry to collect the LMO data displayed in Figure 7. A total of about 5333 data points (30 seconds per point) were employed to span 3.5 to 4.3 V (a half cycle). To remove noise, an 81-point average of the current was implemented, centered about the plotted potential. Last, an ohmic correction was used to align the anodic and cathodic peaks near 4.15 V, and the resulting magnitudes of the anodic and cathodic currents were averaged to obtain a symmetric voltammogram, which was employed in subsequent model-experiment comparison. The specific capacity calculated from the filtered, symmetric voltammogram was 118 mAh/g, consistent with Li-doped LMO ($Li_x[Li_\delta Mn_{2-\delta}]O_4$, with δ being a small number, typically less than 0.2). Equation 5 and the associated text explains how dx/dU can be calculated from the filtered voltammogram.

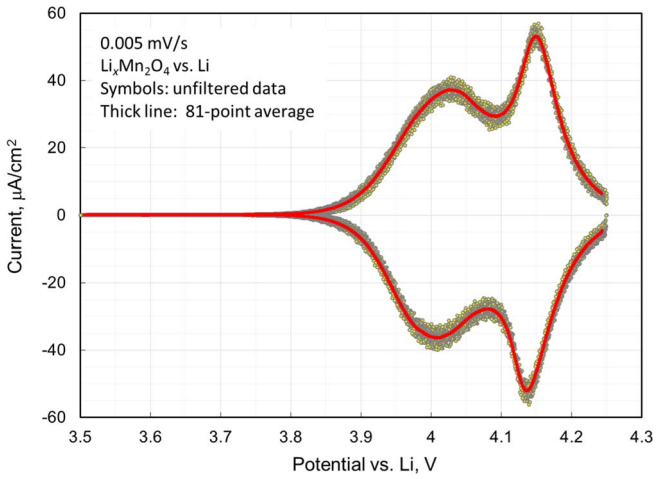

Figure 7. Linear-sweep voltammetry for the $Li_xMn_2O_4$ electrode. To remove noise from the raw data, an 81-point average of the current was implemented, centered about the plotted potential.

Early structural information associated with spinel manganese oxide can be found in the work of Ohzuku et al.[33] Two-phase behavior (cubic phases) are seen for the highest voltage plateau near 4.15 V vs. Li, followed by single-phase behavior upon lithiation to just above 3 V vs. Li. This is consistent with modeling the system with 3 galleries, or $J = 3$, corresponding to the upper plots in Figure 8. Per the nomenclature of Table I, species 2 and 3 make up the chemistry of the uppermost plateau (U^0 values near 4.15 V), and species 1 (with $U_1^0 = 4.01$ V) reflecting the remaining sites, corresponding to a single gallery. The fraction of available total sites X_j compare well with those depicted in Figure 3 of.[33] As defined in the context of the graphitic carbon analysis, the model-experiment comparison is quantitative for $J = 3$. For real materials and electrodes, retention of a purely crystalline, two-phase state is rarely seen upon electrochemical cycling at room temperature. This is likely why we find a good representation of the system can be obtained with $J = 2$ (lower plots of Figure 8), implying that we have difficulty resolving the two cubic phases of the upper voltage plateau, and only one gallery and reaction is needed to obtain a sufficiently accurate representation of both $U(x)$ and dx/dU for upper voltage plateau. We note that a careful comparison and contrasting of the $J = 3$ and $J = 2$ results of the $U(x)$ vs. dx/dU plots (left-most plots of Figure 8) shows that the fit of the upper-most potential peaks is less accurate for $J = 2$ than for $J = 3$. To summarize, as was seen in the previous analyses of Li-graphite, the fit values for the standard electrode potential U_j^0 and the fraction of available sites X_j associated with the reaction of potential U_j^0 are in good agreement with prior X-ray analyses.

Iron phosphate.—GM's first all-electric vehicle after the EV1, which went out of production in 1999, was the 2014 Chevrolet Spark EV, with an iron phosphate positive electrode. The 2014 Spark EV carries an 8-year/100,000-mile warranty. Iron phosphate positives are of current interest for power-battery applications, and single battery solutions with iron phosphate positives have been employed recently for production 12 V start-stop vehicles. The importance of this material has prompted significant research; the positive-electrode reviews mentioned above also cover past developments of lithium iron phosphate.[41–44]

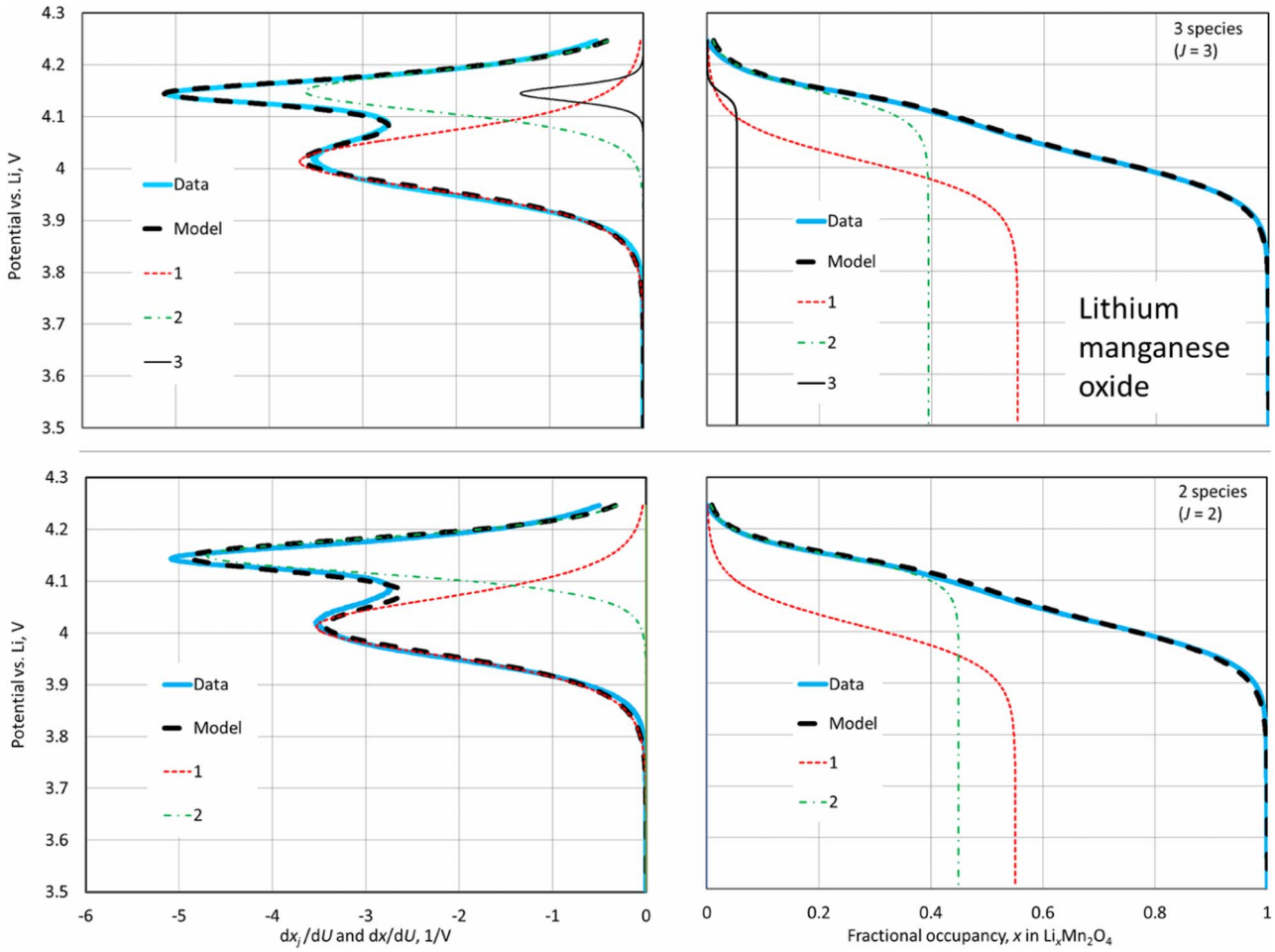

Figure 8. Open-circuit potential (right plots) and differential voltage spectroscopy (left plots) for spinel manganese oxide, including curves for the individual reactions and species. Upper plots: three species. Lower plots: two species.

Application of the model to experimental results for lithium iron phosphate (LFP) is overviewed in Figure 9. Hysteresis is seen in the LFP system.[47–49] To be specific, relative to Figure 9, we see that the potential difference between the oxidation and reduction branches is about 40 mV for $0.1 < x < 0.9$ (with x being defined by Li_xPO_4, and $0 \leq x \leq 1$). This 40 mV can be compared to the potential variation for either the oxidation of reduction potential traces in going from $x = 0.1$ to $x = 0.9$, which is on the order of 1 mV, reflecting near two-phase behavior. In this sense, hysteresis is large in the LFP system, although 40 mV is small relative to Li-graphite/LFP cell potential,

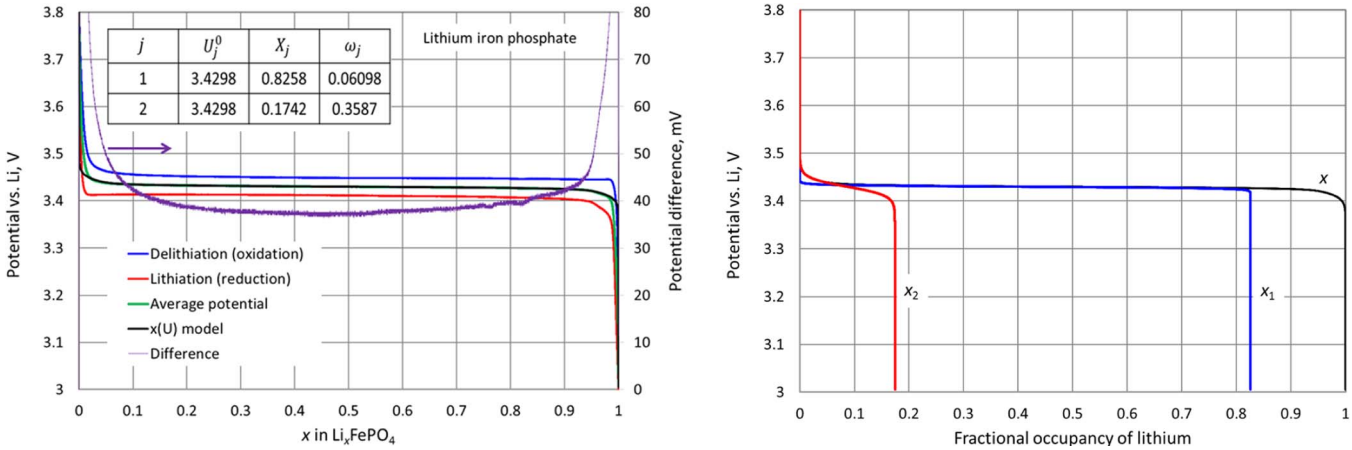

Figure 9. Open-circuit potential (left plot) and potential vs. fractional occupancy for the two species (right plots) for iron phosphate. The green line in the ocp plot is the average of the lithiation (lowermost potential trace) and delithiation (uppermost potential trace) potential curves, and the black line is the model results. The difference between the lithiation and delithiation traces, approximately 40 mV, is shown with the right ordinate (purple curve).

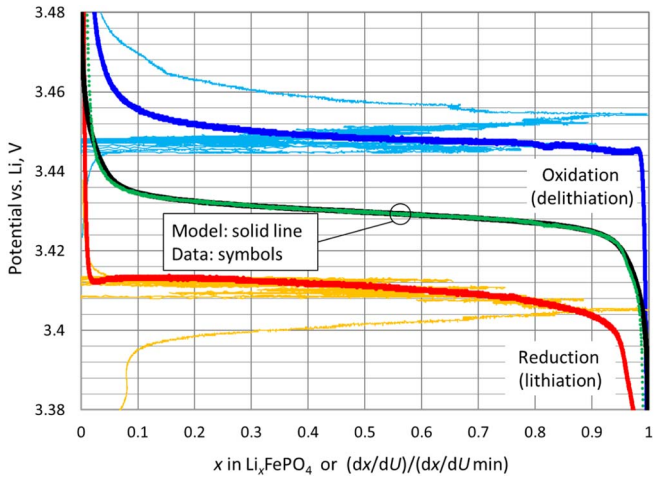

Figure 10. Open-circuit potentials (thicker curves) $U(x)$ and $(dx/dU)/\min(dx/dU)$ (thinner curves) for the lithiation and delithiation potentials. Model: solid black line. Data: green symbols.

which is approximately 3 V, and, for this reason, hysteresis of LFP in full cells is usually not a significant concern. The near two-phase behavior for the majority of the $U(x)$ curves can be attributed to the dominant species ($j = 1$ in the right plot of Figure 9). Curvature near the endpoints of the $U(x)$ plot results from the species $j = 2$, which exhibits a behavior akin to that of a single-phase material (a sloping potential curve). These observations are consistent with prior investigations of Li_xPO_4.[47–49]

While the agreement between the model and the average of the oxidation and reduction ocp's is good, working with the iron phosphate ocp's presents challenges, as can be seen in analyzing Figure 10. Although the ordinate has been expanded relative to Figure 9, it is still difficult to discern the difference between the model (black line) and experimental data (green symbols), and the model-experiment agreement may be viewed as quantitative as defined previously. About 14,000 data points were used to generate each ocp trace; however, determining dx/dU in a manner that enables model-experiment comparison is not achievable using the techniques applied previously in the context of Figure 4, as the numerical differentiation results do not yield sufficient accuracy. As described for the graphite analysis, a least-squares fit of the slope in the $U(x)$ curve—using 501 data points in the iron phosphate analysis—was employed to obtain the dx/dU curves shown in Figure 10 (in light blue and in light orange-yellow colors), which yielded a large amount of noise, albeit less than plots using less data points in the slope analysis. The use of more data points to remove noise in the dx/dU curves that are evident in Figure 10 leads to excessive data loss. Hence, for iron phosphate, we make no attempt to compare model and experiment for dx/dU curves.

Nickel-Manganese-Cobalt oxide.—Variants of $Ni_xMn_yCo_z$ ($x + y + z = 1$), or NMC, are used in GM's Gen 1 and Gen 2 Volt EREV, the Gen 2 Spark EV, and the Bolt EV. Reviews of NMC materials can be found in Refs. 41–44 and 45; original research on the material can be found in Ref. 50–60. In this work, we apply our thermodynamic modeling to $Ni_{0.6}Mn_{0.2}Co_{0.2}$. NMC differs from the other materials studied in this work in that the host material is not stable in conventional applications at oxidizing potentials required to remove all Li (e.g., well above 4.3 V vs. Li). Recent investigations of this aspect of NMC can be found in References 58 and 59; see, in particular, Figure 5 of Ref. 58 and the associated discussion with respect to the inability to remove all Li from NMC at potentials yielding stable performance.

Shown in Figure 11 are the open-circuit potentials $U(x)$ and $(dx/dU)/\max|dx/dU|$ extracted from the experimental data. About 14000 data points were collected to form the $U(x)$ curve, and the 241 points were used in the least-squares fit of the slope of the $U(x)$ curve to extract the $(dx/dU)/\max|dx/dU|$ curves plotted in the left graphic of Figure 11. The averaged potential shown in the right plot corresponds to the average of the oxidation and reduction potentials provided in the left plot. Again, 241 points were used in the least-squares fit of the slope of the $U(x)$ curve of the right plot to formulate the $(dx/dU)/\max|dx/dU|$ of the right plot. The curves shown in the right plot are later discussed in the content of model-experiment comparison. We note that as has been reported previously, two oxidation reactions (at about 3.63 and 3.72 V vs. Li) are seen on oxidation, whereas only the 3.72 V peak is seen on reduction. Hence, the dx/dU curve extracted from the averaged potential (right plot of Figure 11) displays a clear peak at 3.72 V, and a shoulder near 3.63 V. We have not examined this difference in the oxidation and reduction traces further, and it remains an open question.

Slow-sweep-rate cyclic voltammetry results are provided in Figure 12; as expected, the results are very similar to those of the low-current (approximately C/100) results of Figure 11, consistent with the ideal analysis discussed in the context of Figure 2. For sufficiently slow sweep rates, we reach a state of dynamic equilibrium, implying that the $U(x)$ and dx/dU traces do not change as the sweep rate is reduced, indicative of no significant (or measureable) impedances intruding on the traces.

Equations 10 and 12 and the values provided in Table I (for $Li_xNi_{0.6}Mn_{0.2}Co_{0.2}$) are used to calculate the $x(U)$ and dx/dU model curves in Figure 13. Once again, the agreement is quantitative as

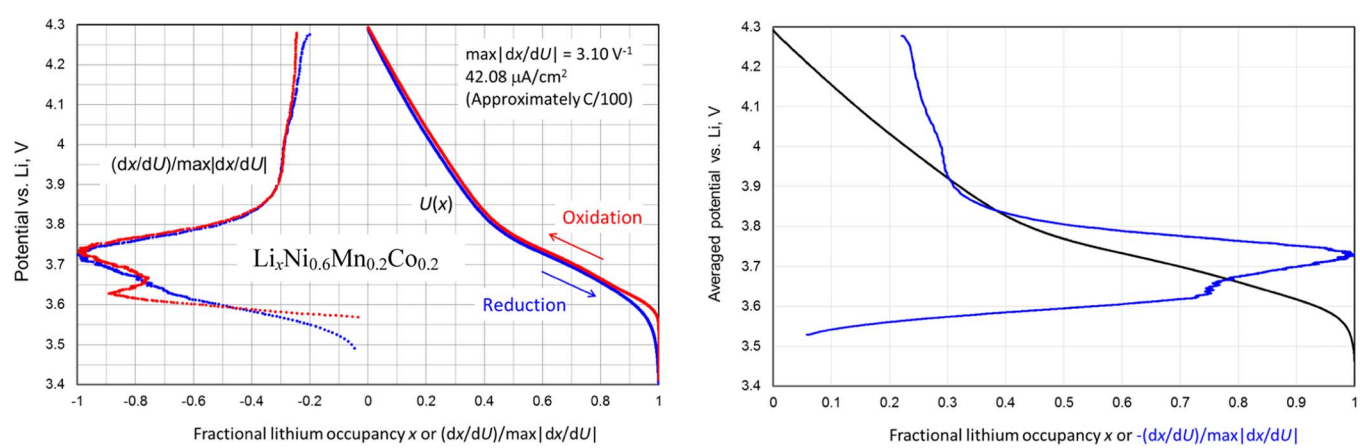

Figure 11. Open-circuit potentials $U(x)$ and $(dx/dU)/\max|dx/dU|$ extracted from the experimental data without further treatment (left plot) and the average of the oxidation (delithiation) and reduction (lithiation) potentials as well as the differential of the average potential curve (right plot) for the $Li_xNi_{0.6}Mn_{0.2}Co_{0.2}$ electrode. The curves in the right plot are used to extract model parameters (cf. Figure 13).

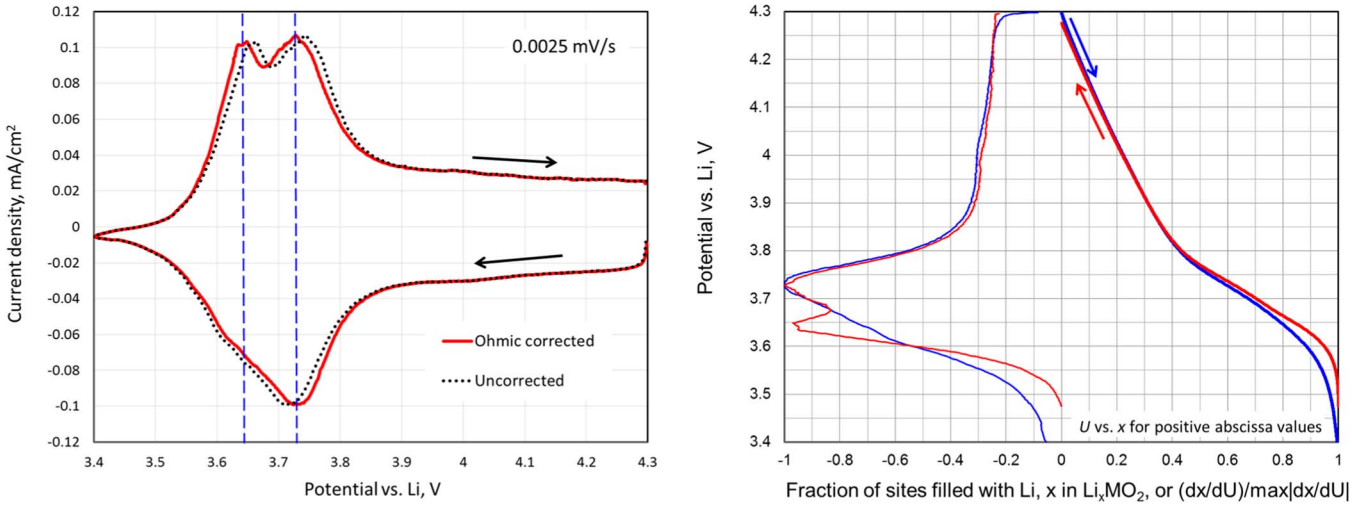

Figure 12. Slow-sweep-rate cyclic voltammetry curves for the $Li_xNi_{0.6}Mn_{0.2}Co_{0.2}$ electrode (left) and differential voltage spectroscopy as well as $U(x)$ curves (right). Results for the third cycle are provided; each half cycle (4.3 to 3.4 V) takes 100 hours to complete. For the red curve on the left and the plot on the right, an ohmic correction to the data of 83.3 Ohm was made to align the peaks at 3.72 V. As with the C/100 results provided in Figure 11, two peaks are seen on oxidation (3.63 and 3.72 V), whereas only the 3.72 V peak shows clearly upon reduction. More generally, the $U(x)$ and $(dx/dU)/\max|dx/dU|$ extracted from both the C/100 and 0.0025 mV/s experimental data are very similar.

defined previously. The electrode is comprised of a number of single-phase galleries and reactions (i.e., $\omega_j \geq 1$ for all four reactions), in agreement with prior literature (for example, see Figure 14 of Ref. 41 and the associated discussion in that publication). The portions of the plots above 4.3 V in Figure 13 of this work are shaded gray to indicate that degradation of the NMC host and solvent oxidation results for such high potentials, and in this sense, these potentials are not of practical interest for this material system.

Conclusions

We describe and implement a thermodynamic model for substitutional electrode materials. To assess the value of the model, we apply it to four insertion-electrode materials of commercial interest today. Specifically:

- An introduction is provided for an ideal reaction system (cf. Eq. 1) to outline our modeling approach and to compare and contrast the merits of constant-current methods vs. those of linear-sweep-voltammetry for dynamic-equilibrium analyses.
- Our previous thermodynamic model that was applied to Li-Si[12] is extended to include reactions between galleries within the electrode materials, and new clarifications are provided based on free energies of the galleries. An appendix is included to address the convergence properties of the governing equilibrium relation (Eq. 10).
- At the close of the Theory section of this paper, we show that if the galleries in the material are assumed to be in local equilibrium, then it is possible to express the chemical potential gradient of lithium in the material such that a single diffusion equation can be employed along with multiple electrochemical reactions with different lithium species at the material surface (cf. Eq. 29).
- Model-experiment comparisons are made for lithiated graphite (Figure 3), manganese oxide (Figure 8), iron phosphate (Figure 9 and Figure 10), and $Li_xNi_{0.6}Mn_{0.2}Co_{0.2}$ (Figure 13). In all cases, the model-experiment comparisons are in quantitative agreement insofar as the differences between the two are less than those associated with

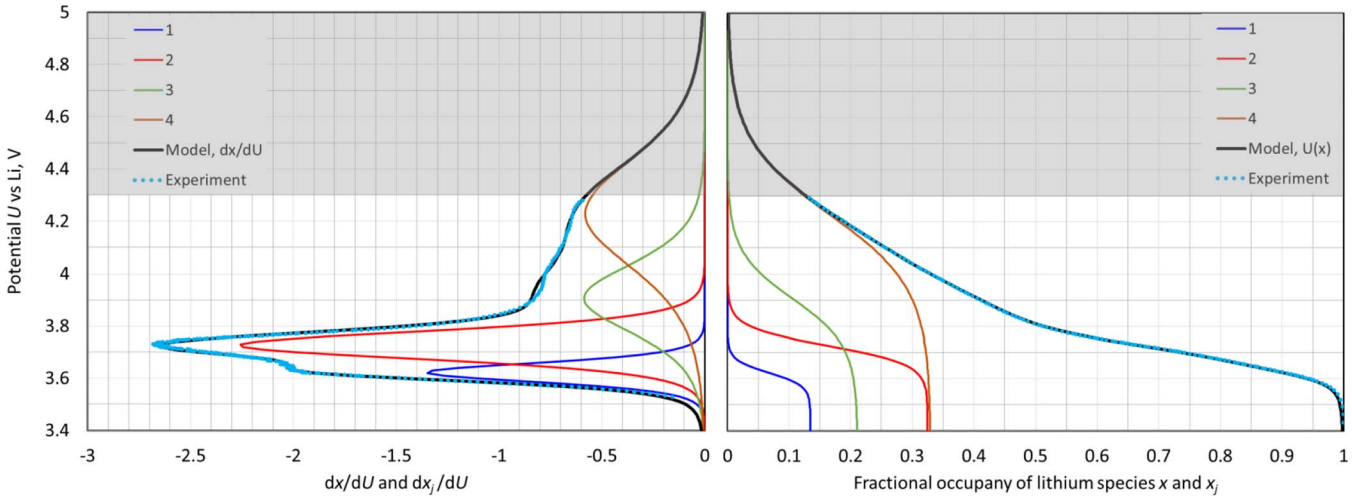

Figure 13. Model-experiment comparison for the open-circuit potential (right) and differential voltage spectroscopy (left) for the $Li_xNi_{0.6}Mn_{0.2}Co_{0.2}$ electrode. The experimental data correspond to the right plot of Figure 11. The model formulation is based on full delithiation taking place at a potential above 4.3 V, which is not accessible experimentally due to electrolyte oxidation and system degradation; hence, the gray shading for potentials above 4.3 V and the use of inaccessible lithium corresponding to a lithium occupancy of $x_0 = 0.1325$ (cf. Table I and associated text).

experiment-to-experiment variation and are difficult to discern in the $U(x)$ plot that spans the minimum and maximum practical values of U and x.

- In all model-experiment comparisons, significant filtering of the data was needed, and the procedures to do this are described. Plots reflecting the filtering treatments for constant-current and linear-sweep-voltammetry experiments are depicted in Figure 4 and Figure 7, respectively.

Acknowledgment

The authors acknowledge helpful discussions with Charles Wampler (GM Research and Development) associated with numerical aspects of this work. We recognize and appreciate the groundbreaking path LG Chem has taken in delivering best-in-class cells to General Motors.

Appendix

While the utility of the proposed model rests in part on the small number of species and reactions j needed to characterize electrode materials and the speciation afforded by the model being in accordance with independent measurements (e.g., X-ray diffraction), it is of practical interest to ascertain whether the series in Eq. 10 converges in terms of replicating $U(x)$ data. That is, upon adding more terms to the series, should we expect greater accuracy in representing the $U(x)$ data? In this Appendix, we will show that any monotone decreasing function $x(U)$ can be represented to arbitrary accuracy as a sum of terms of the form given in Equation 10. To see this, note from Figure 1 that Eq. 10 approaches a step function as ω_j becomes large, with

$$x_j(U) = \begin{cases} 0 \text{ for } U > U_j^0 \\ X_j \text{ for } U < U_j^0 \end{cases}$$

It follows that the sum of any such functions (using large values of ω_j) will resemble a "stair case" where the height of each stair is given as X_j and the placement of each stair is given by the value of U_j^0. Any monotone decreasing function can be represented to arbitrary accuracy by such a "stair case" function, where the accuracy of the approximation increases with the number of stairs that are used in the approximation. This observation suffices to show that any monotone function $x(U)$can be approximated to arbitrary accuracy as a series of terms of the form in Eq. 10.

The above demonstration only requires functions with very large values of ω_j, and no assertions are made about how many such functions would be required to achieve a desired accuracy. As noted above, the real strength of this approximation method seems to lie in the fact that good approximations can be obtained with a small number of such functions, having moderate values of ω_j, and that these functions are in good agreement with X-ray diffraction data in terms of identified phases and species.

References

1. K. E. Thomas, R. M. Darling, and J. Newman, "Mathematical Modeling of Lithium Batteries," in: *Advances in Lithium-Ion Batteries*, W. van Schalkwaijk and B. Scrosati, (eds.), Kluwer, Dordrecht (2002), Chap. 12.
2. J. Newman and K. E. Thomas-Alyea, *Electrochemical Systems*, 3rd Edition, Wiley Interscience, 2004.
3. V. Ramadesigan, P. W. C. Northrop, S. De, S. Santhanagopalan, R. D. Braatz, and V. R. Subramanian, *J. Electrochem. Soc.*, **159**, R31 (2012).
4. K-H. Xue and G. L. Plett, *Electrochim. Acta*, **87**, 575 (2013).
5. A. J. Bard and L. R. Faulkner, *Electrochemical Methods & Applications*, 2nd Edition ed., Wiley, 2000.
6. M. W. Verbrugge and B. J. Koch, *J. Electrochem. Soc.*, **150**, A374 (2003).
7. D. R. Baker and M. W. Verbrugge, *J. Electrochem. Soc.*, **159**, A1341 (2012).
8. R. D. Deshpande, J. Li, Y-T. Cheng, and M. W. Verbrugge, *J. Electrochem. Soc.*, **158**, A845 (2011); M. W. Verbrugge, R. D. Deshpande, J. Li, and Y-T. Cheng, *Proceedings of the Materials Research Society*, Spring Meeting, April 2011.
9. C. R. Birkl, E. McTurk, M. R. Roberts, P. G. Bruce, and D. A. Howey, *J. Electrochem. Soc.*, **162**, A2271 (2015).
10. K. G. Gallagher, D. W. Dees, A. N. Jansen, D. P. Abraham, and S-H. Kang, *J. Electrochem. Soc.*, **159**, A2029 (2012).
11. D. W. Dees, D. P. Abraham, W. Lu, K. G. Gallagher, M. Bettge, and A. N. Jansen, *J. Electrochem. Soc.*, **162**, A559 (2015).
12. M. W. Verbrugge, D. R. Baker, and X. Xiao, *J. Electrochem. Soc.*, **163**, A262 (2016).
13. T. R. Jow and C. C. Liang, *J. Electrochem. Soc.*, **129**, 1429 (1982).
14. E. A. Guggenheim, *Thermodynamics*, 6th ed., North-Holland, Amsterdam, 1977.
15. J. M. Prausnitz, R. N. Lichtenthaler, and E. Gomes de Azevedo, "*Molecular Thermodynamics of Fluid-Phase Equilibria*," 3rd ed., Prentice-Hall, 1999.
16. D. R. Baker, M. W. Verbrugge, and A. F. Bower, *J. Solid State Electrochem.*, **20**, 163 (2016).
17. M. D. Levi, E. A. Levi, and D. Aurbach, *J. Electroanal. Chem.*, **421**, 89 (1997).
18. D. R. Baker, M. W. Verbrugge, and X. Xiao, to be published.
19. T. Ohzuku, Y. Iwakoshi, and K. Sawai, *J. Electrochem. Soc.*, **140**, 2490 (1993).
20. N. Takami, A. Satoh, M. Hara, and T. Ohsaki, *J. Electrochem. Soc.*, **142**, 371 (1995).
21. T. Zheng, J. N. Reimers, and J. R. Dahn, *Phys. Rev. B.*, **51**, 734 (1995).
22. M. D. Levi and D. Aurbach, *J. Electroanal. Chem.*, **421**, 79 (1997).
23. M. D. Levi, E. A. Levi, and D. Aurbach, *J. Electroanal. Chem.*, **421**, 89 (1997).
24. P. Yu, B. N. Popov, and R. E. White, *J. Electrochem. Soc.*, **146**, 8 (1999).
25. M. Winter, J. O. Besenhard, M. E. Spahr, and P. Novák, *Adv. Mater.*, **10**, 725 (1998).
26. C. Lampe-Onnerud, J. Shi, P. Onnerud, R. Chamberlain, and B. Barnett, *J. Power Sources.*, **97–98**, 133 (2001).
27. Y. F. Reyneir, R. Yazami, and B. Fultz, *J. Electroanal. Chem.*, **151**, A422 (2004).
28. J. Christensen and J. Newman, *J. Electrochem. Soc.*, **153**, A1019 (2006).
29. D. K. Karthikeyan, G. Sikha, and R. E. White, *J. Power Sources*, **185**, 1398 (2008).
30. R. A. Huggins, *Advanced Batteries: Materials Science Aspects*, Springer, New York, NY 2009.
31. S. Goriparti, E. Miele, F. De Angelis, E. Di Fabrizio, R. P. Zaccaria, and C. Capiglia, *J. Power Sources*, **257**, 421 (2014).
32. J. C. Hunter, *J. Solid State Chem.*, **39**, 142 (1981).
33. T. Ohzuku, M. Kitagawa, and T. Hirai, *J. Electrochem. Soc.*, **137**, 769 (1990).
34. A. de Kock, M. H. Rossouw, L. A. de Picciotto, M. M. Thackeray, W. I. F Daivid, and R. M. Ibberson, *Mater. Res. Bull.*, **25**, 657 (1990).
35. J-M. Tarascon and D. Guyomard, *J. Electrochem. Soc.*, **138**, 2864 (1991).
36. R. J. Gummow, A. de Kock, and M. M. Thackeray, *Solid State Ionics*, **69**, 59 (1994).
37. G. Pistoia, D. Zane, and Y. Zhang, *J. Electrochem. Soc.*, **142**, 2551 (1995).
38. Y. Gao, J. N. Reimers, and J. R. Dahn, *Phys. Rev. B*, **54**, 3878 (1996).
39. G. G. Amatucci, N. Pereira, T. Zheng, and J-M. Tarascon, *J. Electrochem. Soc.*, **148**, A171 (2001).
40. K. Amine, J. Liu, S. Kang, I. Belharouak, Y. Hyung, D. Vissers, and G. Henriksen, *J. Power Sources*, **129**, 14 (2004).
41. M. S. Whittingham, *Chem. Rev.*, **104**, 4271 (2004).
42. J. W. Fergus, *J. Power Sources*, **195**, 939 (2010).
43. M. M. Doeff, "Battery Cathodes," chapter 2 in *Batteries for Sustainability: Selected Entries from the Encyclopedia of Sustainability Science and Technology*, edited by R. J. Brodd, DOI , Springer, 2013.
44. D. Andre, S-J. Kim, P. Lamp, S. F. Lux, F. Maglia, O. Paschosa, and B. Stiasznya, *J. Mater. Chem. A*, **3**, 6709 (2015).
45. F. Schipper, E. M. Erickson, C. Erk, J-Y. Shin, F. F. Chesneau, and D. Aurbach, *J. Electrochem. Soc.*, **164**, A6220 (2017).
46. A. Bhandari and J. Bhattacharya, *J. Electrochem. Soc.*, **164**, A106 (2017).
47. F. Baronti, N. Femia, R. Saletti, C. Visone, and W. Zamboni, *IEEE Trans. Magnetics*, **50**, 7300704 (2014).
48. W. Dreyer, J. Jamnik, C. Guhlke, R. Huth, Jože Moškon, and M. Gaberšček, *Nature Materials*, "The thermodynamic origin of hysteresis in insertion batteries," **9**, 448 (2010).
49. A. Barai, W. D. Widanage, J. Marco, A. McGordon, P. Jennings, and ",A study of the open circuit voltage characterization technique and hysteresis assessment of lithium-ion cells," *J. Power Sources*, **295**, 99 (2015).
50. N. Yabuuchi and T. Ohzuku, *J. Power Sources*, **119–121**, 171 (2003).
51. J. B. Goodenough and Y. Kim, *Chem. Mater.*, **22**, 587 (2009).
52. M. Shikano, H. Kobayashi, S. Koike, H. Sakaebe, Y. Saito, Hironobu Hori, H. Kageyama, and K. Tatsumi, *J. Power Sources*, **196**, 6881 (2011).
53. H-J. Noh, S. Youn, C. S. Yoon, and Y-K. Sun, *J. Power Sources*, **233**, 121 (2013).
54. J. Li, J. Camardese, S. Glazier, and J. R. Dahn, *Chem. Mater.*, **26**, 7059 (2014).
55. Y. Wei, J. Zheng, S. Cui, X. Song, Y. Su, W. Deng, Z. Wu, X. Wang, W. Wang, M. Rao, Y. Lin, C. Wang, K. Amine, and F. Pan, *J. Am. Chem. Soc.*, **137**, 8364 (2015).
56. J. Li, L. E. Downie, L. Ma, W. Qiu, and J. R. Dahn, *J. Electrochem. Soc.*, **162**, A1401 (2015).
57. R. Amin and Y-M. Chiang, *J. Electrochem. Soc.*, **163**, A1512 (2016).
58. J. Kasnatscheew, U. Rodehorst, B. Streipert, S. Wiemers-Meyer, R. Jakelski, R. Wagner, I. C. Laskovic, and M. Winter, *J. Electrochem. Soc.*, **163**, A2943 (2016).
59. J. Kasnatscheew, M. Evertz, B. Streipert, R. Wagner, R. Klopsch, B. Vortmann, H. Hahn, S. Nowak, M. Amereller, A. C. Gentschev, P. Lamp, and M. Winter, *Phys. Chem. Chem. Phys.*, **18**, 3956 (2016).
60. E. M. Erickson, H. Bouzaglo, H. Sclar, K-J. Park, B-B. Lim, F. Schipper, C. Ghanty, J. Grinblat, B. Markovsky, Y-K. Sun, and D. Aurbach, *J. Electrochem. Soc.*, **163**, A 1348 (2016).